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Multi-objective metaheuristics applied for the multivariable system identification

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ABSTRACT

The mathematical modeling of a multivariable system is a challenging engineering problem for classical mathematical optimization schemes. This study implements and compares multi-objective metaheuristics—Multi-objective Grey Wolf Optimizer (MOGWO), Nondominated Sorting Genetic Algorithm II (NSGA-II), and Multi-objective Cuckoo Search (MOCS)—for the identification of MIMO (Multiple-Input Multiple-Output) systems. Through 100 independent runs, these multi-objective algorithms were evaluated against their single-objective counterparts using performance metrics such as the Coefficient of Determination (R^2) and Mean Squared Error (MSE). The results demonstrate that multi-objective metaheuristics outperform their single-objective variants, proving an effective alternative for accurate and efficient MIMO system identification.

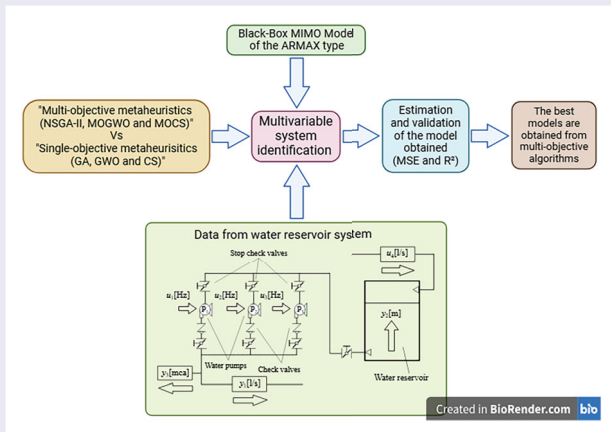
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Multi-objective optimization; metaheuristics; identification system; multivariable systems



1. Introduction

Metaheuristics are often inspired by some behavior in nature, being proposed to solve complex engineering problems. These powerful algorithms are widely applied in challenging optimization problems, where classical methods based on gradient information or mathematical programming perform inefficiently in obtaining acceptable solutions [1, 2].

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Currently, many metaheuristics are being implemented satisfactorily to solve intractable problems, being of the single-objective or multi-objective type. The appeal of using these algorithms to solve complex problems is due to obtaining optimal solutions, even for high-dimensionality problems and in small amounts of time [3].

Metaheuristics inspired by nature can be classified into three main groups based on the theory of evolution, the laws of physics, and swarm intelligence. One can also mention another group of algorithms highlighted as inspired by human behavior [4]. Other authors [5] include a fifth group in this classification called hybrid algorithms. Among the mentioned groups, the theory of evolution and swarm intelligence stands out.

Evolutionary algorithms are population-based and mimic biological evolution. The most popular algorithm in this group is the Genetic Algorithm (GA), introduced by Holland in 1975 [6] and improved by other collaborators [7–9]. This metaheuristic is a mathematical model of biological evolution based on Charles Darwin's theory of natural selection. There is also a multi-objective version of this algorithm called Nondominated Sorting Genetic Algorithm II (NSGA-II), proposed by [10].

The algorithms based on swarm intelligence mimic the social life behavior of certain types of animals. One of the most popular swarm intelligence algorithms is the Grey Wolf Optimizer (GWO) proposed by [11]. This algorithm had inspired by the hunting behavior of grey wolves. GWO was developed for solving global optimization problems. The multi-objective version of this algorithm is called Multi-objective Grey Wolf Optimizer (MOGWO), proposed by [12]. Another metaheuristic of the swarm intelligence group that can cited too is the Cuckoo Search (CS) proposed by [13]. This algorithm models the predatory and parasitic behavior of cuckoos. After, the same authors [14], in another study, proposed a multi-objective extension known as Multi-objective Cuckoo Search (MOCS).

This article aims to apply the single-objective and multi-objective versions of the GA, GWO, and CS in the identification process of the reservoir system. This plant is MIMO (Multiple-Input Multiple-Output) type, being a challenging real multivariable problem for the field of control and automation engineering. Thus, the global optimization capacity of these algorithms was evaluated by computer simulation to highlight the differences between the results obtained by single-objective metaheuristics and their multi-objective versions.

This study evaluates multi-objective metaheuristics and their respective single-objective versions by Coefficient of Determination (R^2) and Mean Square Error (MSE) metrics. This work also analyses the computational cost of all algorithms and their respective results in referent estimation and validation steps of the identification process of a MIMO system.

The original contribution of this work aims to reinforce how advantageous the implementation of multi-objective metaheuristics is in the identification process of MIMO systems. Although single-objective algorithms are less computationally demanding and easier to apply, they require aggregating all system outputs into a single objective function. This study demonstrates that such aggregation can lead to the dominance of one output's optimization over others due to the big difference between the order of magnitude of the outputs, often making it difficult to obtain a desirable solution. In contrast, multi-objective algorithms treat the outputs as individual objectives, where bias error does not affect the quality of the solutions. Under identical experimental conditions (including iteration number and population size), the results confirm that multi-objective metaheuristics consistently outperform their single-objective counterparts.

Despite there are a lot of applications of single-objective and multi-objective metaheuristics in control strategies, as in [15–22], there are not many applications of these algorithms in the mathematical modeling process of industrial systems. In the works that apply metaheuristics in the identification process, it is almost unanimous that the algorithms used are single-objective types, as in [23–25]. The work of [26] is the only one that makes the application of a multi-objective algorithm in the identification process of a MIMO system. It is important to highlight that all real systems have a multivariate nature [27]. These systems are often simplified to a SISO (Single-Input Single-Output) model, where variables (inputs and outputs) of no interest are treated as disturbances or nonlinearities [27]. This affirmation may be one of the justifications for using single-objective algorithms in system identification processes. In contrast, this work is an extension of the study of [26] that aims to show and validate the benefits of applying multi-objective algorithms in the identification process of multivariable systems.

The remainder of this article is structured as follows: In Section 2, the fundamentals of the single-objective metaheuristics and their multi-objective versions were shown respectively; Section 3 initially presents the multivariable system used in the identification process. Next, it shows the criteria of analysis and comparison adopted in this work. This section finishes by showing the results obtained; Finally, Section 4 provides suggestions with constraints and deployment for other authors.

2. Methods

Initially, in this section, it is shown a brief description of the single-objective metaheuristics evaluated in this work. Next, the basis of their respective multi-objective versions is shown.

2.1. Single-objective algorithms

The single-objective metaheuristics used in this work are briefly presented as follows.

2.1.1. GA

This algorithm is a well-known metaheuristic based on evolution theory that applies mathematical operations between binary arrays that mimic the crossover and mutation behaviors of a population of individuals (solutions). In GA, the solutions represent the chromosomes (arrays of bits) of the genetic code. Over generations, solutions evolve through the selection process that evaluates their quality by fitness value from the objective function. Some of the best individuals selected will be used for the next generation. The optimization process finishes when the stop criteria are satisfied.

For GA to find a good or optimal solution, it shows the following steps [28]: (i) encoding the cost functions; (ii) definition of a fitness function or selection criterion; (iii) creation of the population of individuals; (iv) realization of the evolution cycle or iterations by evaluating the fitness of all the individuals in the population, creating a new population by performing crossover and mutation, fitness-proportionate reproduction, etc., and replacing the old population and iterating again using the new population; (v) decodification of the results to obtain the problem solution. These procedures can be seen in pseudocode from Algorithm 1.

Algorithm 1: Pseudocode of the GA [28]

```

1 Define the objective function  $f(x)$ ,  $x = (x_1, \dots, x_d)^T$ ;
2 Encode the solution into chromosomes (binary strings);
3 Define fitness  $F$  (e.g.  $F \propto f(x)$  for maximization);
4 Generate the initial population;
5 Initial probabilities of crossover ( $p_c$ ) and mutation ( $p_m$ );
6 while  $t < \text{maximum number of generations}$  do
7   Generate new solution by crossover and mutation;
8   if ( $p_c > \text{rand}$ ) then
9     Crossover;
10  end
11  if ( $p_m > \text{rand}$ ) then
12    Mutate;
13  end
14  Accept the new solution if its fitness increases;
15  Select the current best for the next generation (elitism);
16 end
17 Finish.

```

The performance of the GA algorithm directly depends on the configuration of its hyperparameters, which are as follows: crossover probability p_c and mutation probability p_m . The population size n , dimension size d , choice of fitness function, and selection criteria of the new population also directly affect the convergence of the GA [28].

The version of GA adopted in this work uses floating point representation because the mathematical modeling problem of the MIMO system has a higher dimensionality. If binary representation were adopted, then the size of the chromosomes could be too large, making the computational cost of this algorithm much higher.

2.1.2. GWO

This algorithm is based on the hunting mechanism of grey wolves, which are natural predators at the top of the food chain. These animals live in groups where there is a dominant social hierarchy. The alpha wolf (α) is responsible for leading the pack of wolves, α represents the best solution. After α in the hierarchy comes the beta wolf (β) and delta wolf (δ) which are the second and the third best solution, respectively. These wolves help the α . The lowest class of the hierarchy is represented by the omega (ω) wolves that submit to the wolves of the upper classes in the search for a global optimum [11, 29].

During the hunting process, the prey stops moving when wolves surround it. The equations representing this behavior are as follows [11]:

$$\vec{D} = |\vec{C} \vec{X}_p(t) - \vec{X}(t)| \quad (1)$$

$$\vec{X}(t+1) = \vec{X}_p(t) - \vec{A} \vec{D} \quad (2)$$

where t indicates the current iteration, $\vec{X}_p(t)$ is the position vector of the best wolf, vector $\vec{D}$ represents the random movement distance of the best wolf to the ω wolf in the current iteration, $\vec{X}(t)$ and $\vec{X}(t+1)$ respectively indicate the position of a ω wolf in the current iteration and the iteration one step forward. The variables $\vec{A}$ and $\vec{C}$ are coefficient vectors, being calculated by the following equations [11]:

$$\vec{A} = 2\vec{a} \vec{r}_1 - \vec{a} \quad (3)$$

$$\vec{C} = 2\vec{\tau}_2 \quad (4)$$

where the elements of $\vec{\tau}$ decrease linearly from 2 to 0 at each iteration. The vectors $\vec{\tau}_1$ and $\vec{\tau}_2$ are randomly generated with uniform distribution in the interval [0, 1].

The vector $\vec{A}$ represents the territorial exploration behavior of the grey wolves and decreases at each iteration because it depends on the $\vec{\tau}$ value. When $|\vec{A}| > 1$, the wolves are exploring the search space (diverging) to find potential prey to chase and attack it. When $|\vec{A}| < 1$, wolves are attacking the prey (converging). Vector $\vec{C}$ simulates the effect of obstacles when wolves are chasing their prey. Unlike vector $|\vec{A}|$, the vector $|\vec{C}|$ acts randomly, facilitating or hindering the capture of the prey [11].

The grey wolves can locate potential prey, then surround and then attack. The position of the ω wolves is updated according to the location of the α , β and δ wolves that lead the hunt, as they have better knowledge about the location of possible prey. The equations that represent this behavior are the following:

$$\vec{D}_\alpha = |\vec{C}_\alpha \vec{X}_\alpha(t) - \vec{X}(t)| \quad (5)$$

$$\vec{D}_\beta = |\vec{C}_\beta \vec{X}_\beta(t) - \vec{X}(t)| \quad (6)$$

$$\vec{D}_\delta = |\vec{C}_\delta \vec{X}_\delta(t) - \vec{X}(t)| \quad (7)$$

$$\vec{X}_1 = \vec{X}_\alpha(t) - \vec{A}_\alpha \vec{D}_\alpha \quad (8)$$

$$\vec{X}_2 = \vec{X}_\beta(t) - \vec{A}_\beta \vec{D}_\beta \quad (9)$$

$$\vec{X}_3 = \vec{X}_\delta(t) - \vec{A}_\delta \vec{D}_\delta \quad (10)$$

$$\vec{X}(t+1) = \frac{\vec{X}_1 + \vec{X}_2 + \vec{X}_3}{3} \quad (11)$$

Equations (5) to (10) represent the implementation of the grey wolf hunting mechanism. These equations are an expansion of (1) to (4), and consider the individual influence of the α , β , and δ wolves. Equation (11) updates the position of the ω wolves for the next iteration, being an average of the $\vec{X}_1$, $\vec{X}_2$, and $\vec{X}_3$ values that respectively represent the individual influence of the α , β , and δ wolves. The ω wolves tend to move towards the center of the top three wolves. At the end of each iteration, the group hierarchy had updated [11]. The pseudocode of the GWO is represented by Algorithm 2.

2.1.3. CS

Cuckoos are known to make beautiful sounds, but some species of these birds (for example: *ani* and *Guira*) have brood parasitism behavior in their reproduction strategy. This behavior happens when female cuckoo lays their eggs in the nests of other host birds (often other species) that just laid their eggs. Some host birds may come into conflict with the intruder cuckoo. When a host bird discovers the eggs are not its own, it can abandon the nest and build another in a new location or expulse the alien eggs. Some species of cuckoos in the Americas (for example: *Tapera*) imitate the color and pattern of the egg from other host birds. This behavior increases the probability of their eggs not being abandoned, and thus also their reproductivity [28].

In addition, generally, the cuckoo's egg hatches a little before the host nest's eggs. After the cuckoo chick is born, the first instinct is to push out the eggs from the host bird.

Algorithm 2: Pseudocode of the GWO [11]

```

1 Define the objective function  $f(X)$ ,  $X = (X_1, \dots, X_n)^T$ ;
2 Initialize the wolves population  $X_i (i = 1, 2, \dots, n)$ ;
3 Initialize the hyperparameters  $\vec{a}$ ,  $\vec{A}$  and  $\vec{C}$ ;
4 Calculate the fitness of each wolf;
5  $X_\alpha$ : the best wolf;
6  $X_\beta$ : the second wolf;
7  $X_\delta$ : the third wolf;
8  $t = 1$ ;
9 while  $t < \text{maximum number of iterations}$  do
10   for each wolf do
11     Update the position of the current wolf by Equation (11);
12   end
13   Update  $\vec{a}$ ,  $\vec{A}$  e  $\vec{C}$ ;
14   Calculate the fitness of all wolves;
15   Update  $X_\alpha$ ,  $X_\beta$  and  $X_\delta$ ;
16    $t = t + 1$ ;
17 end
18 Finish.

```

This behavior eliminates competition for food inside the nest. Studies show that a cuckoo chick can also mimic the call of host chicks to gain access to more feeding opportunities. According to the authors [28], the CS follows three basic rules:

- Each cuckoo lays one egg at a time, and dumps its egg in a randomly chosen nest.
- The best nests with high-quality eggs will be carried over to the next generations.
- The number of available host nests n is fixed, and the egg laid by a cuckoo is discovered by the host bird with a probability $p_a \in [0, 1]$. In this case, the host bird can either get rid of the egg or simply abandon the nest and build a completely new nest.

The CS algorithm uses a mechanism that updates new solutions based on Lévy flights [30]. In nature, many animals and insects show this behavior. For generating a new solution x^{t+1} , the equation that uses the Lévy flight is calculated by:

$$x_i^{t+1} = x_i^t + \alpha \oplus \text{Lévy}(\lambda) \quad (12)$$

where $\alpha > 0$ is the step size and λ is Lévy exponent. The entrywise product $\oplus$ is applied in these values to update the solution x^t . The Lévy flight function represents a random walk whose a random step length. The equation that represents this behavior is approximately calculated by:

$$\text{Lévy}(\lambda) \sim e^{-\lambda}, \quad (1 < \lambda \leq 3) \quad (13)$$

The pseudocode of the CS is represented by Algorithm 3.

2.2. Multi-objective algorithms

The multi-objective versions of the GA, GWO, and CS algorithms are as follows.

Algorithm 3: Pseudocode of the CS [28]

```

1 Define the objective function  $f(x)$ ,  $x = (x_1, \dots, x_d)^T$ ;
2 Generate initial population of  $n$  host nests  $x_i$ ;
3 while  $t < \text{maximum number of generations}$  do
4   Get a cuckoo randomly;
5   Generate a solution by Lévy flights by Equation (12);
6   Evaluate its solution quality or objective value  $f_i$ ;
7   Choose a nest among  $n$  (say,  $j$ ) randomly;
8   if  $f_i < f_j$  then
9     Replace  $j$  by the new solution  $i$ ;
10  end
11  A fraction ( $p_a$ ) of worse nests are abandoned;
12  New nests/solutions are built/generated;
13  Keep best solutions (or nests with quality solutions);
14  Rank the solutions and find the current best;
15   $t = t + 1$ ;
16 end
17 Finish.

```

2.2.1. NSGA-II

One of the well-known multi-objective versions of the GA algorithm is the NSGA-II proposed by [10]. This metaheuristic is one of the most popular of the group of multi-objective evolutionary algorithms. NSGA-II was proposed to alleviate three shortcomings in multi-objective algorithms that use nondominated sorting and sharing: computational complexity, nonelitism approach, and the need for specifying a sharing parameter [10]. NSGA-II is a version improved of the nondominated sorting genetic algorithm (NSGA) of [31] that has these three deficiencies.

NSGA-II has a scheme of new population selection based on two operators: fast non-dominated sorting and a metric for management of the population diversity called crowding distance. The non-dominated sorting mechanism divides the population into non-dominated levels (fronts) since the initialization process of the algorithm. The first level corresponds to a non-dominated solutions set. The solutions from the second level are dominated by solutions of the first front and simultaneously dominate the next level. Similarly, at other levels, the solutions of the respective previous levels dominate. In population, each individual has its rank value associated with a particular front. Adding, the crowding distance is computed too. The best solutions with lower ranks or greater crowding distance are on the first front. In the NSGA-II, parents' solutions are selected first by the best of rank and crowding distance values, made by binary tournament selection. Next, crossover and mutation operators are applied in these solutions to generate offspring that will be sorted based on non-domination together with the current population. Finally, this process updates the population and will repeat interactively until a stop criterion is satisfied. The pseudocode of the NSGA-II is represented by Algorithm 4.

2.2.2. MOGWO

The multi-objective version of the GWO was proposed by [12]. The difference between GWO and MOGWO is that the multi-objective version has two additional mechanisms responsible for exploring solutions in a multi-objective search space. The first mechanism consists of an archive responsible for storing non-dominated Pareto optimal solutions. The second

Algorithm 4: Pseudocode of the NSGA-II [32]

```

1 Define the number of functions  $m$  for the multi-objective problem;
2 Initialize objective functions  $f_1(x), \dots, f_m(x)$ ,  $x = (x_1, \dots, x_d)^T$ ;
3 Initialize the population  $x_i (i = 1, 2, \dots, d)$ ;
4 Evaluate objective values;
5 Assign Rank (level) Based on Pareto dominance - sort;
6 Generate Child Population;
7 Binary Tournament Selection;
8 Recombination and Mutation;
9 for  $i$  to  $g$  do
10   for each Parent and Child in Population do
11     Assign Rank (level) based on Pareto - sort;
12     Generate sets of nondominated vectors along PF known;
13     Loop (inside) by adding solutions to next generation starting from the first front until N individuals found
       determine crowding distance between points on each front;
14   end
15   Select points (elitist) on the lower front (with lower rank) and are outside a crowding distance;
16   Create next generation;
17   Binary Tournament Selection;
18   Recombination and Mutation;
19 end
20 Finish.

```

mechanism is a leader selection strategy that helps select the α , β , and δ wolves from the solutions saved in the archive [12].

The archive is a simple storage unit that can save or retrieve non-dominated Pareto optimal solutions obtained during the iterations. This unit has a controller that controls when a solution wants to enter the archive or when the same is full of non-dominated solutions. The archive has a limit to the number of non-dominated Pareto optimal solutions that can store. During the iterations, the non-dominated solutions obtained are compared with the resident solutions from the archive. The results of these comparisons can be as follows [12]:

- If a new solution was dominated by at least one of the archive's resident solutions, then this new solution must not be allowed to enter the archive.
- If a new solution dominates one or more solutions residing in the archive, then dominated solutions must be omitted, and the new solution will be able for enter the archive.
- If neither the new solution or the archive's resident solutions dominate each other, then new solution must be added to the archive.
- If the archive is full, then a grid mechanism must run first to reorganize the target space segmentation and find the most crowded segment to omit one of its solutions. Next, new solution must be inserted to the least crowded segment. This mechanism aims to improve the diversity of the final approximated Pareto optimal front.

The probability of excluding a non-dominated solution is increased proportionally to the number of solutions in the hypercube (segment). For removing solutions if the archive is full, the most crowded segments are first selected. Next, a solution is omitted from one of them randomly to provide a space for the new solution. There is a special case where a

solution is inserted outside the hypercubes. In this case, all segments are extended to cover the new solutions. Therefore, segments of other solutions can also be changed [12].

In [12], the leader selection mechanism chooses the least crowded segments in the multi-objective search space and offers one of its non-dominated solutions as α , β and δ wolves. The roulette wheel selection method is used to choose the leading wolves using the following probability for each hypercube:

$$p_i = \frac{c}{N_i} \quad (14)$$

where c is a constant greater than 1 and N_i corresponds to the number of Pareto optimal solutions obtained in the i th segment. The pseudocode referring to this metaheuristic is represented by Algorithm 5.

Algorithm 5: Pseudocode of the MOGWO [12]

```

1 Define the number of functions  $m$  for the multi-objective problem;
2 Initialize objective functions  $f_1(X), \dots, f_m(X)$ ,  $X = (X_1, \dots, X_n)^T$ ;
3 Initialize the wolves population  $X_i (i = 1, 2, \dots, n)$ ;
4 Initialize the hyperparameters  $\vec{d}$ ,  $\vec{A}$  and  $\vec{C}$ ;
5 Calculate the objective values of each wolf;
6 Find the non-dominated solutions and initialize the archive with them;
7 Select  $X_\alpha$  by leader selection strategy mechanism;
8 Exclude  $\alpha$  from the archive to avoid selecting the same leader;
9 Select  $X_\beta$  by leader selection strategy mechanism;
10 Exclude  $\beta$  from the archive to avoid selecting the same leader;
11 Select  $X_\delta$  by leader selection strategy mechanism;
12 Add back  $\alpha$  and  $\beta$  to the archive;
13  $t = 1$ ;
14 while  $t < \text{maximum number of iterations}$  do
15   for each wolf do
16     | Update the position of the current wolf by Equation (11);
17   end
18   Update  $\vec{d}$ ,  $\vec{A} \in \vec{C}$ ;
19   Calculate the objective values of all wolves;
20   Find the non-dominated solutions;
21   Update the archive with respect to the obtained non-dominated solutions;
22   if the archive is full then
23     | Run grid mechanism to omit one of the current archive members;
24     | Add the new solution to the archive;
25   end
26   if any of the new added solutions to the archive is located outside the hypercubes then
27     | Update the grids to cover the new solution(s);
28   end
29   Select  $X_\alpha$  by leader selection strategy mechanism;
30   Exclude  $\alpha$  from the archive to avoid selecting the same leader;
31   Select  $X_\beta$  by leader selection strategy mechanism;
32   Exclude  $\beta$  from the archive to avoid selecting the same leader;
33   Select  $X_\delta$  by leader selection strategy mechanism;
34   Add back  $\alpha$  and  $\beta$  to the archive;
35    $t = t + 1$ ;
36 end
37 Finish.

```

2.2.3. MOCS

The MOCS algorithm proposed by [14] has two modifications in the first and third of the three basic rules in the single objective version. These reformulations are necessary to solve

multi-objective optimization problems with K different objectives. According to [14], the first and third rules modified and incorporated are follows as:

- Each cuckoo lays K eggs at a time, and dumps them in a randomly chosen nest. Egg k corresponds to the solution to the k th objective.
- Each nest will be abandoned with a probability p_a and a new nest with K eggs will be built, according to the similarities or differences of the eggs. Some random mixing can be used to generate diversity.

MOCS uses the same strategies as the NSGA-II to obtain the current Pareto optimal solutions. The fast non-dominated sorting and crowding distance approaches make this metaheuristic more effective in solving multi-objective problems [33]. The pseudocode of the MOCS is represented by Algorithm 6.

Algorithm 6: Pseudocode of the MOCS [14]

```

1 Define the number of functions  $m$  for the multi-objective problem;
2 Initialize objective functions  $f_1(x), \dots, f_m(x)$ ,  $x = (x_1, \dots, x_d)^T$ ;
3 Generate an initial population of  $n$  host nests  $x_i$  and each with  $K$  eggs;
4 while ( $t < \text{maximum number of generations}$ ) bfor (stop criterion) do
5   Get a cuckoo (say  $i$ ) randomly by Lévy flights;
6   Evaluate and check if it is Pareto optimal;
7   Choose a nest among  $n$  (say  $j$ ) randomly;
8   Evaluate  $K$  solutions for nest  $j$ ;
9   if new solutions of nest  $j$  dominate those of nest  $i$  then
10    | Replace nest  $i$  by the new solution set of nest  $j$ ;
11   end
12   Abandon a fraction ( $p_a$ ) of the worse nests;
13   Keep the best solutions (or nests with non-dominated sets);
14   Sort and find the current Pareto optimal solutions;
15 end
16 Finish.

```

3. Results

This work evaluates the single-objective metaheuristics and their multi-objective versions in the identification process of a multivariable system of the MIMO type, analyzing the differences between the algorithms in solving a real problem. Mathematical modeling of a MIMO system is challenging because it contains many parameters to estimate and validate a model that represents the behavior of a complex system.

3.1. Description of the multivariable system

The multivariate system used to analyze the algorithms consists of four inputs and three outputs. The MIMO system is a water reservoir system, as shown in Figure 1. One of the four inputs is the input flow that feeds the system, while the other three inputs are the frequency value of the inverters that control the water pumps that empty the reservoir. One of the three outputs is the level of the water reservoir, while the other two are the flow and the

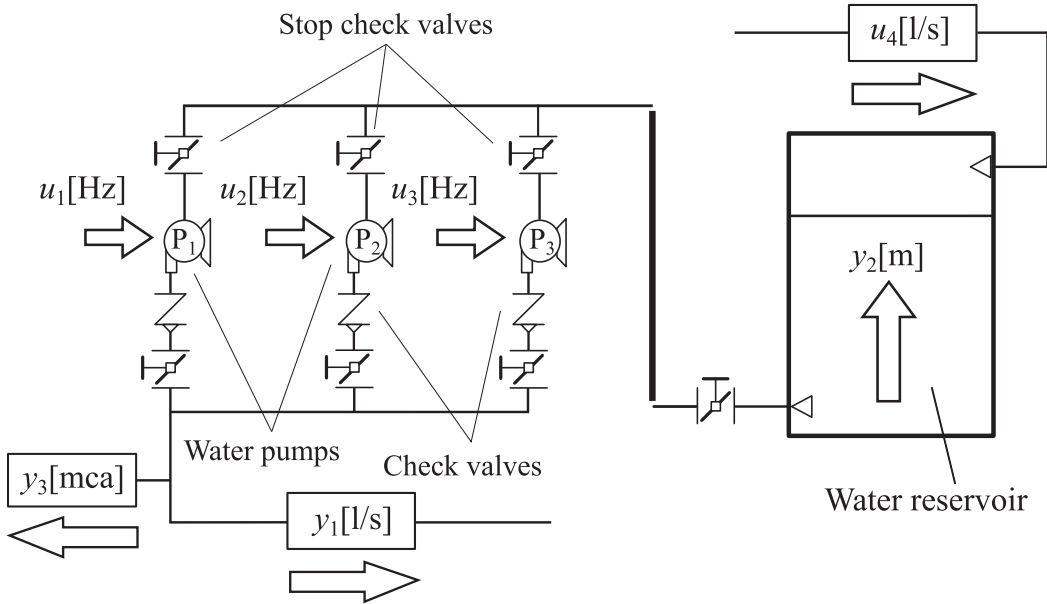


Figure 1. The scheme of the water reservoir system.

pressure concerning the volume of water supplied by the three water pumps (P_1 , P_2 , and P_3).

For the identification process, multivariable systems had considered challenging to obtain an adequate mathematical model. This is due to the variation in the behavior of these systems that may depend on the simultaneous interaction of their variables [27].

Figure 2 shows the data used in the estimation and validation step of identification process of the mathematical model in (15). The estimation step uses a sample of 600 elements, while the validation uses 500. Each element contains values of four inputs and three outputs of a given time. These data have a sampling time equal to one hour. These data had provided by a sanitation company located in Curitiba city, Parana state, Brazil [34]. For the identification process, it hadn't applied any preprocess method to this data.

3.2. Description of the mathematical model

The identification process of this work had made by predicting the error, and the mathematical model adopted is of the black box type. This type is used when only the inputs and outputs data of the system are known [35].

The mathematical model used in the identification process is a system of three polynomial equations represented in discrete time k with a sampling time equal to one hour. This mathematical equation is an Autoregressive Moving Average with eXogenous inputs (ARMAX) model, where it was obtained empirically.

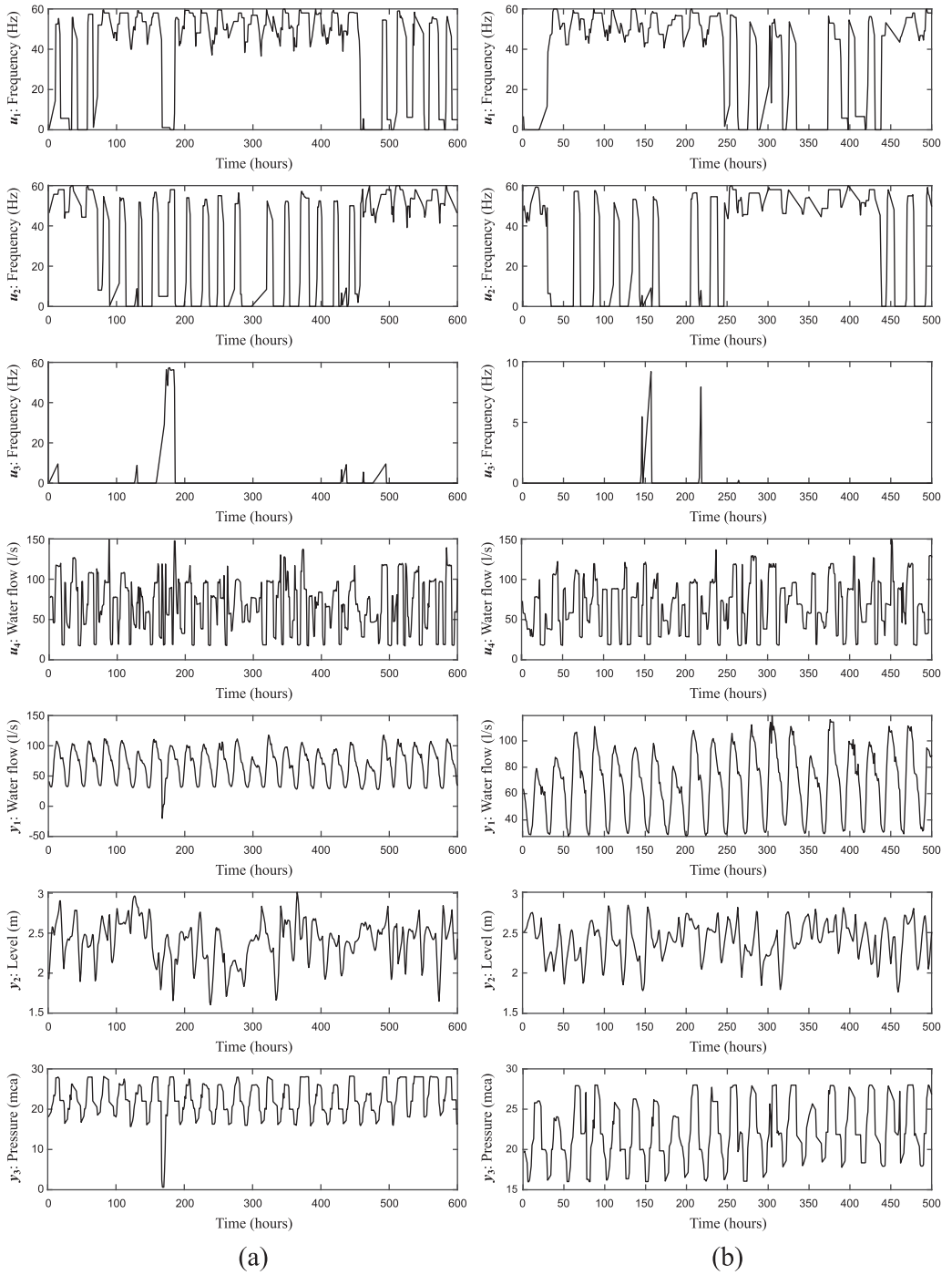


Figure 2. Data from water reservoir system: (a) estimation and (b) validation.

The ARMAX model used in the identification is:

$$\begin{aligned}
 y_1[k] = & -a_{11}y_1[k-1] - a_{12}y_1[k-2] - a_{13}y_1[k-3] - a_{112}y_2[k-1] - a_{122}y_2[k-2] \\
 & - a_{132}y_2[k-3] - a_{113}y_3[k-1] - a_{123}y_3[k-2] - a_{133}y_3[k-3] + b_{111}u_1[k-1] \\
 & + b_{121}u_1[k-2] + b_{131}u_1[k-3] + b_{112}u_2[k-1] + b_{122}u_2[k-2] + b_{132}u_2[k-3] \\
 & + b_{113}u_3[k-1] + b_{123}u_3[k-2] + b_{133}u_3[k-3] + b_{114}u_4[k-1] + b_{124}u_4[k-2] \\
 & + b_{134}u_4[k-3] + c_{11}e_1[k-1] + c_{12}e_1[k-2] + c_{13}e_1[k-3] \\
 y_2[k] = & -a_{21}y_2[k-1] - a_{22}y_2[k-2] - a_{23}y_2[k-3] - a_{211}y_1[k-1] - a_{221}y_1[k-2] \\
 & - a_{231}y_1[k-3] - a_{213}y_3[k-1] - a_{223}y_3[k-2] - a_{233}y_3[k-3] + b_{211}u_1[k-1] \\
 & + b_{221}u_1[k-2] + b_{231}u_1[k-3] + b_{212}u_2[k-1] + b_{222}u_2[k-2] + b_{232}u_2[k-3] \\
 & + b_{213}u_3[k-1] + b_{223}u_3[k-2] + b_{233}u_3[k-3] + b_{214}u_4[k-1] + b_{224}u_4[k-2] \\
 & + b_{234}u_4[k-3] + c_{21}e_2[k-1] + c_{22}e_2[k-2] + c_{23}e_2[k-3] \\
 y_3[k] = & -a_{31}y_3[k-1] - a_{32}y_3[k-2] - a_{33}y_3[k-3] - a_{311}y_1[k-1] - a_{321}y_1[k-2] \\
 & - a_{331}y_1[k-3] - a_{312}y_2[k-1] - a_{322}y_2[k-2] - a_{332}y_2[k-3] + b_{311}u_1[k-1] \\
 & + b_{321}u_1[k-2] + b_{331}u_1[k-3] + b_{312}u_2[k-1] + b_{322}u_2[k-2] + b_{332}u_2[k-3] \\
 & + b_{313}u_3[k-1] + b_{323}u_3[k-2] + b_{333}u_3[k-3] + b_{314}u_4[k-1] + b_{324}u_4[k-2] \\
 & + b_{334}u_4[k-3] + c_{31}e_3[k-1] + c_{32}e_3[k-2] + c_{33}e_3[k-3]
 \end{aligned} \tag{15}$$

Equation (15) was employed to formulate the objective function, where the parameters a_{ij} , a_{ijk} , b_{ijk} , and c_{ij} (indexes: $i = 1$ to 3 ; $j = 1$ to 3 and $k = 1$ to 4) are estimated. Inputs are u_1 , u_2 , u_3 , and u_4 , while outputs are y_1 , y_2 , and y_3 . The variables e_1 , e_2 , and e_3 are, respectively, the prediction errors of outputs y_1 , y_2 , and y_3 .

3.3. Analysis and comparison criteria

In this study, all algorithms were submitted to 100 runs to evaluate their convergence. In the identification process, referring to the estimation and validation stages, the MSE and R^2 were adopted as criteria to evaluate the quality of the mathematical models obtained from each metaheuristic. These metrics are well-known in the literature, and R^2 is more informative in the evaluation of math models than other metrics such as symmetric mean absolute percentage error (SMAPE), mean absolute error (MAE), mean absolute percentage error (MAPE), MSE, and root mean squared error (RMSE) [36]. For stopping criteria, it adopted a maximum number of 1000 iterations. The values obtained from the identification process are represented by mean μ and standard deviation σ . For multi-objective algorithms, these statistical values were obtained from the result of the Pareto Front of all 100 runs. The identification process in this work is a maximization problem, where the R^2 value evaluates the result of each algorithm. The conventional formula of the R^2 evaluates each objective of the multi-objective algorithms in the estimation and validation stages. For the single-objective metaheuristics, the multiple determination coefficient (R_n^2) transforms the MIMO system into a single-objective problem. This metric was also used to select the best solution from the Pareto Front of all 100 runs for each multi-objective metaheuristic.

The conventional and multiple versions of the equations of MSE and R^2 are as follows:

$$MSE = \frac{1}{N} \sum_{k=1}^N [y(k) - \hat{y}(k)]^2 \quad (16)$$

$$R^2 = 1 - \frac{\sum_{k=1}^N [y(k) - \hat{y}(k)]^2}{\sum_{k=1}^N [y(k) - \bar{y}]^2} \quad (17)$$

$$MSE_n = \frac{1}{n.N} \sum_{i=1}^n \sum_{k=1}^N [y_i(k) - \hat{y}_i(k)]^2 \quad (18)$$

$$R_n^2 = 1 - \frac{\sum_{i=1}^n \sum_{k=1}^N [y_i(k) - \hat{y}_i(k)]^2}{\sum_{i=1}^n \sum_{k=1}^N [y_i(k) - \bar{y}_i]^2} \quad (19)$$

where i and k are respectively indexes corresponding to the i th output and k th sample of the process $y_i(k)$ being analyzed, n corresponds to the output number, $\bar{y}_i$ is the average of the N samples measured for the output i of the process, and $\hat{y}_i$ is the output forecast of the process. When the R^2 value is equal to 1, it indicates an exact adequacy of the model for the measured data of the process. The value of R^2 between 0.9 and 1 is considered sufficient for practical control applications [37]. These conditions also apply to R_n^2 values. Equations (18) and (19) were extended from discussions from works [38–40] that approach multivariable regression.

The population size of GA, GWO, and CS equals 200, 800, and 100, respectively. Despite the differences, the multi-objective versions have the same size, making a comparison as fair as possible between a single-objective metaheuristic and the corresponding multi-objective version. These values were adjusted empirically to obtain the best results from each algorithm (single-objective and multi-objective) with the smallest possible population size. The search space comprises the interval $[-1, 1]$. Table 1 represents the parameters of all algorithms evaluated in the identification process. In this work, GWO and MOGWO parameters from Table 1 are the same as used in [26]. The parameters of the other algorithms were also empirically adjusted based on the performance, where each pair (single-objective and their multi-objective version) uses the same parameter values in the common to have a fairer comparison.

The implementation of this study was done through MATLAB's computational environment (version 2021b in operational system Microsoft Windows 10 on a Core i7 processor with 3.2 GHz and 16 GB of random access memory). Codes of all algorithms used in the identification process are available in [34], together with data from the water reservoir system.

3.4. Results obtained

The numerics results obtained in the 100 runs by all the metaheuristics used in this study are in Tables 2 and 3, while graphics results of process identification are in Figures 3–5. The multi-objective algorithms proved more effective in modeling the MIMO system.

In Table 2, multi-objective algorithms showed better results than their single-objective versions during the estimation and validation steps of the parameters obtained from the ARMAX model with MIMO structure (15). Despite the higher computational cost of the

Table 1. Parameters values of algorithms.

Algorithms	Parameters	Values
GA	Crossover percentage p_c	0.4
	Mutation percentage p_m	0.8
NSGA-II	Crossover percentage p_c	0.4
	Mutation percentage p_m	0.8
GWO	Control parameter $\vec{a}$	[2,0]
MOGWO	Random parameter $\vec{C}$	[0,2]
	Control parameter $\vec{a}$	[2,0]
	Random parameter $\vec{C}$	[0,2]
	Archive size	30
	Grid inflation parameter	0.1
	Number of grids per each dimension	10
	Leader selection pressure parameter	4
	Repository member selection pressure	2
	Discovery rate of alien eggs p_a	0.25
CS	step size α	0.01
	Lévy exponent λ	1.5
	Discovery rate of alien eggs p_a	0.25
MOCS	step size α	0.01
	Lévy exponent λ	1.5
	Number of eggs layed by each cuckoo K	3

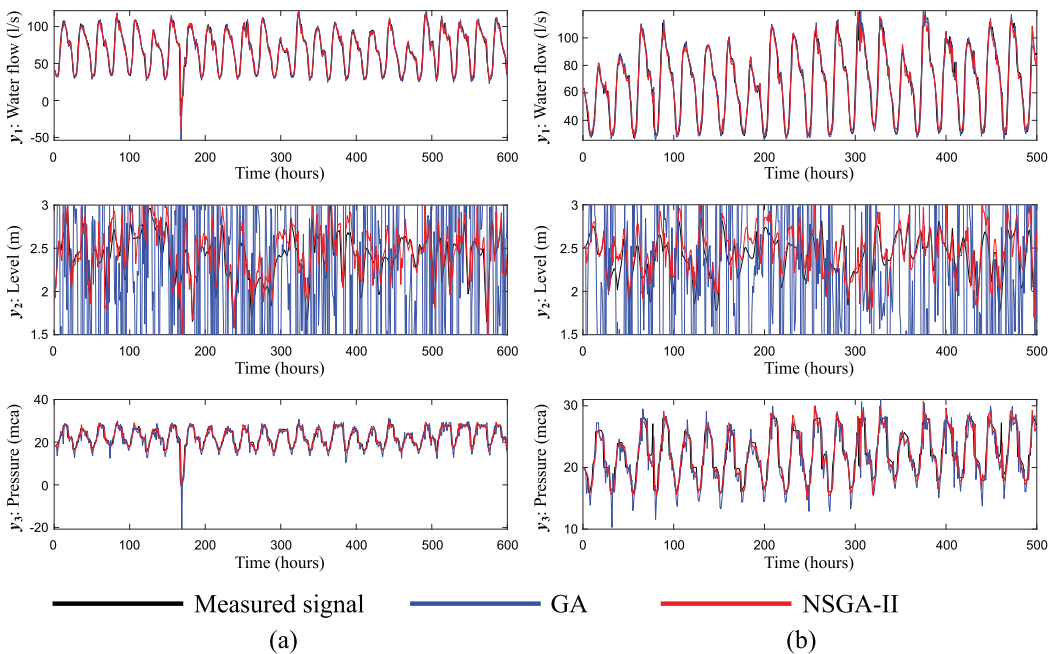


Figure 3. Result of the identification process of GA and NSGA-II algorithms: (a) estimation and (b) validation.

multi-objective metaheuristics used in this study, MOCS and MOGWO showed a lower average value MSE and higher R^2 in most outputs (y_2 and y_3) of the MIMO ARMAX model obtained when compared with their respectively single-objective versions. MOCS showed the best results (result values in bold), while MOGWO was the second (result values in italics). Already, NSGA-II was superior to GA in the modeling of all outputs. This standard

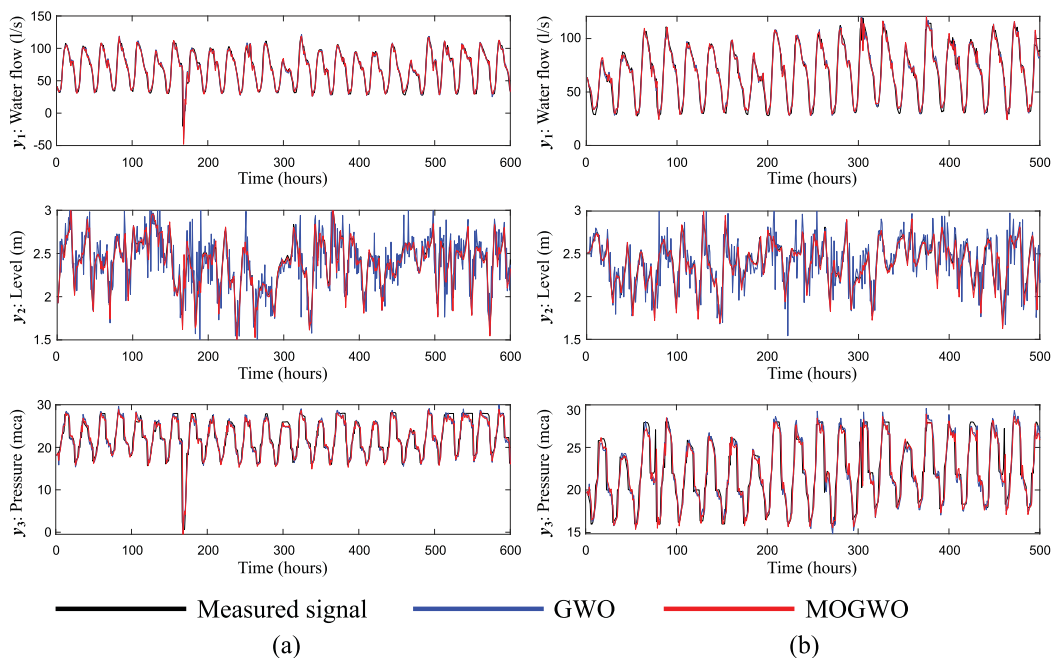


Figure 4. Result of the identification process of GWO and MOGWO algorithms: (a) estimation and (b) validation.

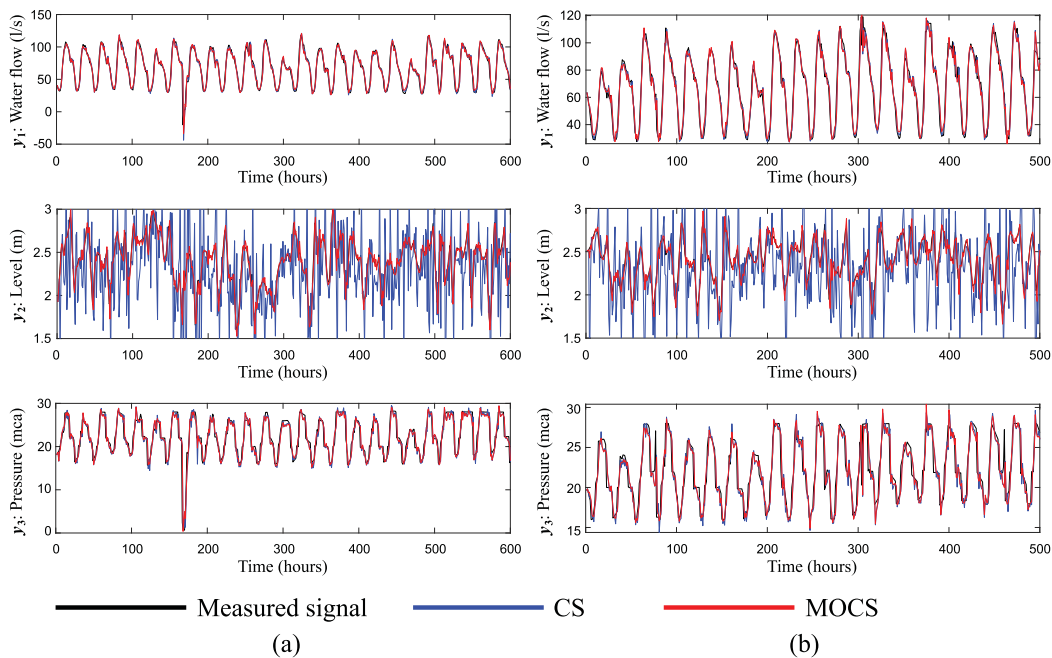


Figure 5. Result of the identification process of CS and MOCS algorithms: (a) estimation and (b) validation.

Table 2. Performance of the single-objective and multi-objective algorithms.

Data obtained			Metaheuristics					
			GA	NSGA-II	GWO	MOGWO	CS	MOCS
MSE_3 (Estimation)	μ		22.070	16.255	15.735	16.622	15.591	15.868
		σ	1.1923	0.5199	0.6139	0.0536	0.1689	0.7318
MSE_3 (Validation)	μ		17.682	12.935	13.048	13.374	12.562	12.845
		σ	1.1433	0.3466	0.5544	0.0536	0.1847	0.4598
R_3^2 (Estimation)	μ		0.9059	0.9307	0.9329	0.9291	0.9335	0.9323
		σ	0.0051	0.0022	0.0026	2.29e-4	7.21e-4	0.0031
R_3^2 (Validation)	μ		0.9190	0.9408	0.9403	0.9388	0.9425	0.9412
		σ	0.0052	0.0016	0.0025	2.50e-4	8.46e-4	0.0021
MSE (Estimation)	y_1		54.933	45.988	41.033	44.158	43.256	44.998
		σ	1.9491	1.5278	1.8812	0.1605	0.4239	2.1925
	y_2		3.8857	0.0167	0.0483	0.0046	0.3215	0.0045
		σ	1.9591	0.1075	0.0252	3.67e-5	0.1114	0.0040
	y_3		7.3922	2.7602	2.9997	2.8269	3.1965	2.6010
		σ	1.8043	0.3688	0.3767	0.0039	0.1843	0.0748
MSE (Validation)	y_1		43.034	36.212	36.460	37.606	34.523	36.164
		σ	2.2080	0.9895	1.6563	0.1636	0.5083	1.3700
	y_2		3.3751	0.0155	0.0416	0.0047	0.2640	0.0039
		σ	1.7948	0.1049	0.0232	3.91e-5	0.1061	0.0034
	y_3		6.6368	2.5766	2.6419	2.5111	2.8984	2.3675
		σ	1.7350	0.3535	0.2824	0.0048	0.2161	0.0775
R^2 (Estimation)	y_1		0.9197	0.9328	0.9355	0.9313	0.9368	0.9342
		σ	0.0028	0.0022	0.0027	2.35e-4	0.0006	0.0032
	y_2		-54.890	0.7598	0.3056	0.9338	-3.6245	0.9358
		σ	28.179	1.5464	0.3626	5.28e-4	1.6028	0.0576
	y_3		0.6124	0.8553	0.8427	0.8518	0.8324	0.8636
		σ	0.0946	0.0193	0.0198	2.06e-4	0.0097	0.0039
R^2 (Validation)	y_1		0.9329	0.9435	0.9431	0.9413	0.9461	0.9436
		σ	0.0034	0.0015	0.0026	2.55e-4	0.0008	0.0021
	y_2		-62.844	0.7061	0.2139	0.9114	-3.9948	0.9257
		σ	33.951	1.9853	0.4379	7.41e-4	2.0067	0.0636
	y_3		0.5314	0.8181	0.8135	0.8227	0.7954	0.8328
		σ	0.1225	0.0250	0.0199	3.41e-4	0.0153	0.0055
Time (s)	μ	47.324	2150.6	28.964	440.19	16.239	32.239	
	σ	1.0146	7.1911	0.4557	5.6440	0.1801	1.1888	
MSE_3 (The best model from validation)			16.473	13.131	11.732	11.935	12.543	11.818
R_3^2 (The best model from validation)			0.9246	0.9399	0.9463	0.9454	0.9426	0.9460
MSE (The best model from validation)	y_1		43.303	37.087	32.738	33.486	34.885	33.082
	y_2		0.8913	0.0027	0.0311	0.0025	0.2289	0.0029
	y_3		5.2259	2.3028	2.4275	2.3168	2.5168	2.3694
R^2 (The best model from validation)	y_1		0.9324	0.9421	0.9489	0.9478	0.9456	0.9394
	y_2		-15.860	0.9495	0.4108	0.9526	-3.3297	0.9596
	y_3		0.6310	0.8374	0.8286	0.8364	0.8223	0.8633

also repeats when comparing the values MSE and R^2 of the best results obtained from the validation process by each pair of algorithms (a single-objective metaheuristic and its multi-objective version).

Seeing Table 2, MOCS also obtained the best MIMO ARMAX model of the identification process. The version multi-objective of the CS shows a computational cost not much higher than the CS, near GWO, lower than GA, and much lower than NSGA-II and MOGWO. This metaheuristic required a much smaller population size to obtain similar results about MOGWO, which justifies the brutal differences in computational cost between the two metaheuristics.

The results obtained by metrics used in multiple regression (MSE_3 and R_3^2 , seeing Table 2) indicate mistakenly that single-objective algorithms as GWO and CS are better than their

Table 3. The best model obtained of the water reservoir system.

Parameter	Metaheuristics					
	GA	NSGA-II	GWO	MOGWO	CS	MOCS
a_{11}	-0.415	-0.580	-1.249	-1.205	-1.000	-1.000
a_{12}	-0.333	-0.119	-7.4e-4	-0.011	-0.187	-0.262
a_{13}	-5.1e-3	-0.036	0.385	0.397	0.519	0.581
a_{112}	-0.226	-0.994	-0.492	-4.168	-0.654	-0.915
a_{122}	-0.626	-0.995	-7.015	0.218	-0.774	-0.985
a_{132}	0.481	-1.000	0.284	0.041	-0.979	-0.969
a_{113}	-0.251	0.487	0.606	-0.036	0.355	0.895
a_{123}	-0.200	-0.384	-7.6e-3	-0.063	-0.207	-0.497
a_{133}	-0.201	-0.998	-0.013	6.0e-3	-0.821	-0.994
b_{111}	0.092	-2.8e-4	-2.7e-4	2.3e-5	0.013	0.039
b_{121}	-0.105	5.2e-4	-1.6e-4	-0.034	0.015	0.020
b_{131}	0.130	-0.046	0.078	0.062	0.020	-0.015
b_{112}	0.037	0.032	-2.4e-4	0.012	0.060	0.049
b_{122}	9.8e-3	0.014	0.024	-2.9e-3	0.011	0.049
b_{132}	0.027	-0.089	0.070	0.040	-0.016	-0.050
b_{113}	0.121	0.010	0.055	0.013	-0.030	-0.011
b_{123}	-0.014	-0.013	-0.002	-6.9e-3	0.022	-0.037
b_{133}	0.080	0.077	4.5e-4	-7.8e-3	0.122	0.163
b_{114}	-0.085	-0.045	-0.013	-0.018	-0.051	-0.046
b_{124}	-0.021	-0.023	4.9e-4	-0.002	0.014	-3.7e-3
b_{134}	-0.028	-0.023	-3.1e-4	-2.6e-3	8.4e-3	0.021
c_{11}	0.763	0.867	0.024	0.034	0.275	0.314
c_{12}	0.145	0.105	7.7e-3	0.012	-0.013	-0.083
c_{13}	-0.129	-0.189	0.169	-6.3e-3	0.276	0.289
a_{21}	-0.021	-0.261	-0.901	-0.650	0.078	0.647
a_{22}	-0.132	-0.222	4.2e-3	-0.296	-0.098	-0.383
a_{23}	0.067	-0.162	0.007	-0.068	0.139	-0.936
a_{211}	-0.041	3.0e-3	1.5e-3	4.5e-3	0.018	0.010
a_{221}	0.093	3.9e-3	7.5e-4	-2.6e-3	1.4e-3	7.2e-5
a_{231}	-0.025	2.8e-3	-8.2e-5	1.2e-3	-4.9e-3	2.4e-3
a_{213}	0.226	-8.3e-3	-0.012	1.0e-3	-0.025	-0.045
a_{223}	0.031	-0.012	0.026	1.6e-3	-4.0e-3	0.014
a_{233}	-0.138	-0.015	4.6e-4	2.0e-3	-0.026	-0.016
b_{211}	0.029	6.1e-4	-1.8e-3	4.9e-4	0.021	-1.1e-3
b_{221}	0.036	-1.6e-4	2.0e-5	-1.1e-3	0.012	8.7e-4
b_{231}	0.026	-1.1e-3	-7.7e-4	-7.8e-5	-0.016	-7.0e-4
b_{212}	4.8e-3	4.7e-4	0.055	-6.9e-4	0.033	-3.3e-4
b_{222}	0.048	7.1e-5	1.3e-3	-9.5e-4	-0.010	-6.9e-4
b_{232}	0.020	-6.2e-4	2.2e-3	1.3e-3	-0.029	1.4e-4
b_{213}	-0.153	2.9e-3	3.4e-4	5.0e-3	-0.014	0.011
b_{223}	0.043	-1.4e-3	-0.011	-3.5e-3	-0.022	-0.010
b_{233}	0.168	-2.8e-3	0.025	-2.2e-3	0.012	1.9e-3
b_{214}	8.6e-3	2.5e-3	3.2e-3	3.7e-3	7.9e-4	3.7e-3
b_{224}	-0.016	1.7e-3	4.6e-3	3.5e-4	1.2e-3	3.5e-3
b_{234}	-1.1e-3	7.3e-4	-2.2e-4	4.7e-4	6.9e-3	2.1e-3
c_{21}	0.767	0.756	1.683	-0.014	0.732	0.931
c_{22}	-0.138	0.134	-0.818	0.235	0.163	0.352
c_{23}	0.226	0.105	-0.018	3.1e-3	0.052	-0.390
a_{31}	-0.159	-0.286	0.076	-0.150	-0.332	-0.262
a_{32}	-0.081	-0.106	-1.2e-3	-0.092	-0.084	0.083
a_{33}	-0.144	-0.134	1.4e-4	-0.196	-0.207	-0.395
a_{311}	0.040	-0.081	-0.100	-0.099	-0.043	-0.071
a_{321}	0.013	-0.010	-0.036	-0.017	-0.035	-0.021
a_{331}	-0.037	0.039	-3.7e-3	0.037	0.054	0.055
a_{312}	0.305	-0.681	-6.502	-3.475	-0.590	-0.817

(continued).

Table 3. Continued.

Parameter	Metaheuristics					
	GA	NSGA-II	GWO	MOGWO	CS	MOCS
a_{322}	-0.241	-0.989	0.733	<i>9.7e-3</i>	-0.546	-0.998
a_{332}	-0.288	-0.947	0.023	<i>-0.011</i>	-0.996	-0.996
b_{311}	0.111	0.041	0.018	<i>-2.4e-3</i>	0.045	0.028
b_{321}	-0.016	-0.011	1.4e-6	<i>8.4e-4</i>	-0.011	-2.6e-3
b_{331}	-0.052	-6.6e-3	5.9e-3	<i>-0.010</i>	3.4e-3	-0.016
b_{312}	0.118	0.027	-1.4e-3	<i>-1.6e-3</i>	0.022	0.017
b_{322}	-0.037	-9.4e-4	3.4e-3	<i>-3.2e-3</i>	0.024	7.1e-3
b_{332}	0.055	-4.5e-3	7.9e-3	<i>-2.8e-3</i>	-0.013	-0.015
b_{313}	0.314	0.070	9.9e-4	<i>-0.010</i>	0.074	0.031
b_{323}	-0.294	-0.043	2.3e-3	<i>0.032</i>	1.5e-3	4.0e-3
b_{333}	0.183	6.1e-3	4.2e-3	<i>-0.023</i>	-0.038	-2.2e-3
b_{314}	-0.021	-2.0e-3	2.6e-3	<i>-1.8e-3</i>	-7.1e-3	-4.0e-3
b_{324}	1.9e-3	2.9e-3	1.9e-4	<i>1.1e-4</i>	-7.0e-3	1.3e-3
b_{334}	2.9e-3	-0.013	-0.011	<i>-6.8e-3</i>	4.0e-4	-4.5e-3
c_{31}	0.570	0.283	0.721	<i>0.540</i>	0.387	0.336
c_{32}	0.259	0.122	-1.0e-4	<i>4.6e-3</i>	0.055	0.236
c_{33}	0.144	0.128	-8.1e-5	<i>-0.012</i>	0.237	-0.096

multi-objective versions, without considering the best model result from the validation process, where MOCS was better than CS. This error happened because there is a big difference between the order of magnitude of the MIMO outputs, thus biasing the average values of MSE_3 and R_3^2 of the GWO and CS. These algorithms were only better than their multi-objective version in modeling the output y_1 (variable of greater magnitude). This result was dominant and contaminated the values of the MSE_3 and R_3^2 metrics. Moreover, GA, GWO, and CS metaheuristics cannot handle objectives individually. Applications of some preprocess methods before the identification process can improve the performance of single-objective metaheuristics used in this work, thus minimizing the effect caused by the big difference between the order of magnitude of the MIMO outputs.

Table 3 shows the parameters of the best MIMO ARMAX model obtained from each metaheuristic used in the identification process of the water reservoir system. These parameters generated the curves of estimation and validation of the Figures 3–5. Note that the results of the MOCS are in bold, indicating the best model obtained. Already, the values of the MOGWO are in italics, indicating the second-best model obtained.

Figures 3–5 confirm the results obtained from Table 2 (estimation and validation step). Unlike multi-objective metaheuristics (curve in red), single-objective algorithms (curve in blue) didn't show good results in modeling the output y_2 of the water reservoir system, with R^2 not near the values range recommended by [37]. Therefore, the best results presented by NSGA-II, MOGWO, and MOCS can be used as a mathematical model that represents well the dynamics of the MIMO system evaluated in this study.

4. Conclusion

This work is an extension of the study [26], where two different multi-objective metaheuristics are included in an identification process of a multivariable system too. These algorithms were compared with their single-objective versions, thus validating and reinforcing the results obtained from the study of [26].

The multi-objective metaheuristics proved effective in the identification process of a MIMO system used in this study. This result is due to the selection method of these algorithms, which are based on Pareto dominance. NSGA-II, MOGWO, and MOCS showed superior to their single-objective versions. Unlike single-objective, multi-objective algorithms do not need a preprocessing method for application before the identification process of any multivariable system.

Although the computational cost is higher when compared to their single-objective version, this work reinforces how beneficial and advantageous is the implementation of a multi-objective algorithm in the identification process of a multivariable system can be. Note that MOCS showed a computational cost that was not much higher than other single-objective algorithms. Then computational cost can not be a serious problem, depending on the multi-objective metaheuristic chosen.

For future works, other researchers can also explore other types of multi-objective metaheuristics that include algorithms with a different selection approach than the method based on Pareto dominance. This work can encourage the implementation of multi-objective metaheuristics in other challenging problems that have not been applied in engineering or other areas yet.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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